Bellhop: the robot that answers the call bell

Oct 9, 2026 · @yash

Bellhop is a small AI robot that rolls to a resident's room when the bell rings, talks to them, and calls a nurse only when the situation needs one.

The problem

Care homes are short-staffed, and one nurse can be responsible for 20, 30 or even 50 residents. Residents are told to use the call bell only for emergencies or something important, but some ring to talk or to ask for a coffee. A nurse cannot tell from the bell alone which kind of call it is, so every ring pulls them away from someone who may need clinical care.

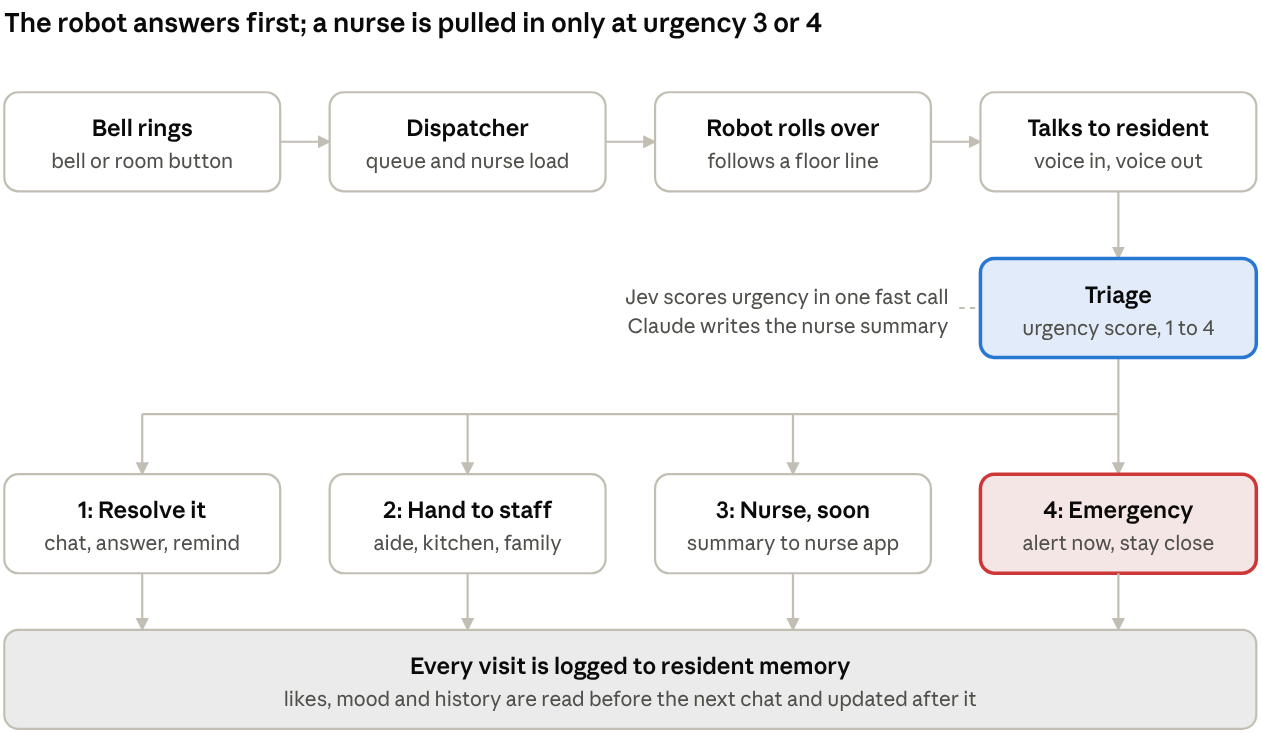
The numbers behind it (sources and caveats are on the Research tab):

- **Thin staffing.** No federal staffing ratio is in force. US nursing homes average 3.68 nursing hours per resident per day against 4.10 recommended, and 70.3% fall short of the 2.45 aide-hour benchmark. In four New York homes each aide had 8 to 13 residents and each LPN 9 to 14.
- **Calls that don't need a nurse.** In a survey of 808 hospital staff only 52% said calls needed a nurse, with about 6.5 calls an hour per person at roughly 4 minutes each. By our estimate that is about 1.7 hours of every 8-hour shift. In one nursing home, 72% of staff who answered open questions said residents ring for non-urgent requests.
- **Slow responses.** In a 300-bed home the average response was 9 minutes, about half of calls missed the home's 5-minute standard, and in 7.5% of calls residents got up or shouted for help.
- **Falls are the danger.** More than half of nursing-home residents fall each year, and in well-staffed homes each extra RN hour went with nearly 40% fewer injurious falls.

We found no study that counts deaths caused by non-urgent calls, so our claims are about nurse time and response speed, not lives saved.

Pitch: don't send a nurse to every bell. Send a robot first, and let it work out whether a nurse is needed.

How it works



bell to outcome · 4 steps, 4 urgency levels

Bellhop reads the resident's memory before it speaks, so it already knows who it is talking to. Urgency 1 and 2 never reach a nurse, 3 arrives with a short written summary, and 4 sets off an alert immediately while the robot stays with the resident.

The agents

Six core agents, each with one job, pass work to each other, and a seventh is a stretch goal. The split also makes the build easy to share across teammates.

Agent	Job	Runs on	Uses
Dispatcher	Decides who answers each bell: robot, nurse, or both, using the resident's risk flags and how busy nurses are	Cloud	Shared realtime state

Agent	Job	Runs on	Uses
Navigator	Drives to the room, stops for obstacles, parks facing the resident	Robot (Pi 5)	Line tracking, ultrasonic sensor, servo head
Companion	Holds the conversation: listens, answers, jokes, reminds, stays calm with confused residents	Cloud + robot audio	Speech-to-text, Claude, text-to-speech
Triage	Scores urgency 1 to 4 from what was said, how it was said and what the camera sees	Cloud	Jev for the fast score, Claude for the reasoning
Memory	Keeps each resident's profile: likes, people, routines, mood over time. Fed to Companion before every visit	Cloud	Vector search, resident database
Staff Liaison	Pings the right person with a short summary, or creates a task for an aide, the kitchen or family	Cloud	Nurse dashboard, push alerts
Courier (stretch)	Carries a nurse-loaded medication tray and opens it only after a nurse approves	Robot	Servo-locked tray, approval from the dashboard

Hardware and materials

The robot is the Freenove 4WD Smart Car Kit with a Raspberry Pi 5 on top. The kit already gives us four drive wheels, a camera and ultrasonic head on servos, line-tracking sensors and LEDs. We add audio, a bell and a few props.

Part	Role in Bellhop	Status
Freenove 4WD Smart Car Kit	Chassis, motors, camera, ultrasonic sensor, servo head, line tracking, LEDs	Have

Part	Role in Bellhop	Status
Raspberry Pi 5	Runs the robot's drive code, camera and audio streaming	Have
Batteries for kit and Pi	Kit does not include them; use a separate stable supply for the Pi so motors cannot brown it out	Confirm
USB speakerphone (mic + speaker)	The robot's ears and voice; the kit has no audio	Confirm
Wi-Fi bell button (ESP32 or Pico W)	The call bell; pressing it sends a call to the Dispatcher. A web button works as a backup	Confirm
Black tape and QR stickers	Hallway line for Navigator; one QR per room door so the camera knows where it is	Make
Small screen or tablet on the robot	Shows captions for hard-of-hearing residents and a friendly face (optional)	Optional
Servo-latched tray	Medication carrier for the Courier stretch goal, filled with fake pills	Stretch
Laptop or tablet	Nurse dashboard and alerts during the demo	Have
Phone hotspot or travel router	Backup network in case venue Wi-Fi is unreliable	Confirm

The kit's LEDs double as a status light: idle, listening, nurse on the way and emergency each get a colour, so judges can read the robot's state from across the room.

Software and tools

Most of this is glue around services we call, which is what makes 48 hours realistic. Claude writes the glue; we spend our time on the robot and the demo.

Layer	Choice	Why
Robot code	Python on the Pi 5, starting from Freenove's open-source car server	Motors, servos, camera and sensors already work

Layer	Choice	Why
Voice	Cartesia Ink (speech-to-text) and Sonic (text-to-speech), streamed. Piper running on the Pi is the local option, with voice picks on the Research tab	Fast first audio and built-in turn detection, so it feels like a conversation
Conversation brain	Claude with a per-resident persona and memory in the prompt	Warm, flexible dialogue and good judgement on edge cases
Urgency scoring	Jev, with Claude as the fallback and the explainer (details on the Research tab)	One fast typed call returns urgency and category with probabilities
Agent orchestration	Claude Agent SDK for Companion and Triage; plain async Python workers for the rest	Keeps each agent small and testable
Shared realtime state	SpacetimeDB: call queue, robot status, resident profiles, visit log	The dashboard updates live with no polling code
Memory recall	Semantic search over each resident's notes, from the sponsor list below	Pulls the right memory into the chat in milliseconds
Nurse dashboard	Web app (TypeScript/React) with a live call feed, robot status, summaries and a nurse-minutes-saved counter	The screen judges will look at
Alerts	Dashboard banner plus a phone notification	Urgency 3 and 4 must reach a human fast
Vision (stretch)	Claude vision on camera snapshots for "person on the floor" and posture checks	Adds an emergency trigger beyond what the resident says

Sponsor APIs for bonus points

Bonus points are optional, so we pick a few sponsors that do real work in the build rather than name-dropping ten. The first five rows are the recommended set; each has a genuine job in the flow above. What each sponsor offers and how sure we are is on the Research

tab.

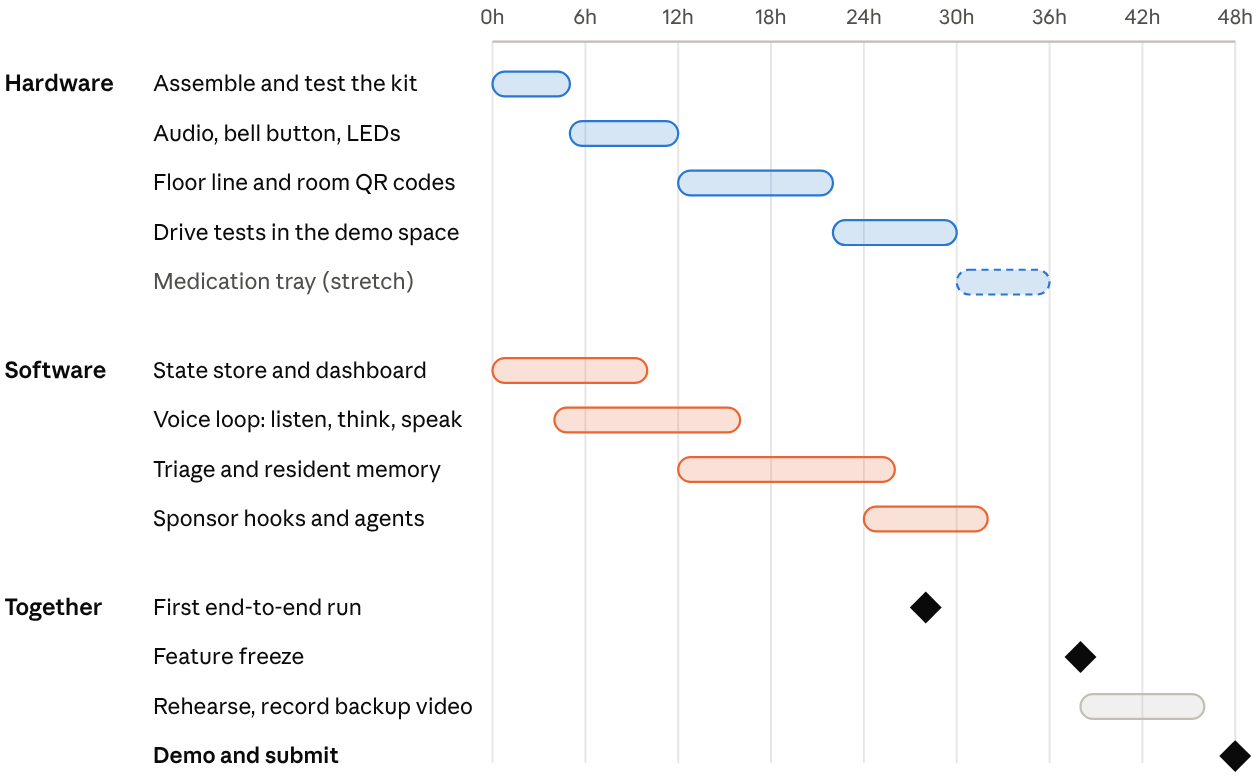
Sponsor	Job in Bellhop	Fit
Anthropic (cohost)	Claude is the Companion's brain, writes the nurse summaries, reasons about edge cases and reads camera snapshots	Core
Cartesia	Streaming speech-to-text with turn detection and fast text-to-speech, so the robot talks like a person	Core
SpacetimeDB (cohost)	Shared realtime state for the call queue, robot status, resident profiles and visit log; the dashboard updates live	Core
Fetch.ai	Register Dispatcher, Triage and Liaison as agents on Agentverse so a nurse can ask "what happened in room 12?" through ASI:One	Strong
Moss	Sub-10 ms semantic recall of a resident's preferences and history, inside the voice loop	Strong
Engramme	Alternative or add-on memory layer for long-term resident memory, if API access works	Optional
Elastic	Hybrid search over care-plan notes and visit logs; Agent Builder for a nurse-facing question agent	Optional
Convex	Alternative to SpacetimeDB for shared state; we use one or the other, not both	Optional
SpaceXAI	Grok Voice as a multilingual voice fallback if Cartesia has an outage	Optional
Runpod	GPU endpoint for an open vision model (fall detection) or an offline-capable Llama if we want a privacy story	Stretch
Context.dev	Pulls today's weather, news or sports so the robot has something fresh to chat about	Stretch
Creao	Builds a scheduled shift-summary agent for nurses without extra code	Stretch

Sponsor	Job in Bellhop	Fit
Coinbase	x402 payments let an agent reorder pills and supplies in stablecoin after a nurse approves (inventory idea on the Extra ideas tab)	Stretch

48-hour build plan

Hardware and software run in parallel, with a first end-to-end run at hour 28

48-hour plan, hours from the start of the hackathon



48-hour plan · 3 lanes, 13 items

The first end-to-end run (a bell press that ends in a spoken answer and a dashboard entry) is the milestone that matters. Everything after it is improvement, and the last ten hours are for rehearsal, not new features.

Risks and fallbacks

Risk	Fallback
Venue Wi-Fi is slow or drops, and voice needs the network	Phone hotspot or travel router; keep replies short so audio streams quickly
Floor navigation is unreliable	Drive the robot with the Freenove app for the demo while everything else stays real
Pi browns out when the motors draw current	Separate stable supply for the Pi, as in the kit's own reviews
Triage gets a case wrong on stage	Bias toward escalating; Claude double-checks any score below a confidence threshold; rehearse scripted cases
A sponsor API has no access or breaks	Each sponsor sits behind one small adapter, so Claude-only or local versions can replace it in minutes

Demo script

Three short scenes, one per outcome tier, with a teammate playing the resident. The dashboard stays on screen throughout.

Scene	Resident says	Bellhop does	Dashboard shows
1. Just lonely	"Can I get a cup of tea and a chat?"	Rolls over, greets her by name, remembers she takes two sugars and asks about her grandson, and sends a tea request to the kitchen	Urgency 1, resolved, no nurse alert
2. Needs a nurse, not urgently	"My hip has been aching since last night."	Asks where and how bad on a 1 to 10 scale, reassures her, and sends the nurse a short summary	Urgency 3 card with the summary; nurse taps "on my way"

Scene	Resident says	Bellhop does	Dashboard shows
3. Emergency	"I've fallen and I can't get up." (or the camera sees someone on the floor)	LEDs go red, alerts the nurse immediately, stays and keeps the resident calm until help arrives	Urgency 4 banner, alert time, live transcript

The closing shot is the counter for calls the robot resolved and nurse minutes saved. It uses an assumed minutes-per-call that we state openly on screen.

Alternative opening: judges play the residents

- Bellhop opens by saying who it is and what it can do, which steers judges toward questions we have rehearsed.
- A teammate stands behind the robot holding funny prompt cards, such as "ASK IT TO PLAY A GAME", to guide the next question.
- This runs for about the first minute. Then we switch to the dashboard and show that the conversation was recorded and the judge is now a resident in the database.
- The venue will be loud, so the judge has to speak close to the robot or it may pick up someone else. A small, close-up robot makes that feel natural.

Cost and savings

A robot is a one-time hardware cost plus usage-based API fees, so extending to more residents means adding robots. In the base case one robot frees about 970 staff hours a year, worth about $24,100 at a loaded $24.80 an hour, for about $3,200 a year in running costs, and the hardware pays back in under two months. These are model results from untested assumptions (4 minutes saved per call, a $2,500 fleet robot, 40 calls resolved a day), laid out with sources on the Research tab. Freed time becomes cash only if it cuts overtime or agency hours, or buys more clinical care.

Calls resolved a day	Staff hours freed a year	Equivalent staff	Value of freed time	API plus upkeep	Net a year	Hardware payback
10	243	0.12	$6,035	$1,087	$4,948	6.1 months
20	487	0.23	$12,069	$1,799	$10,270	2.9 months
40 (base)	973	0.47	$24,139	$3,222	$20,917	1.4 months

Calls resolved a day	Staff hours freed a year	Equivalent staff	Value of freed time	API plus upkeep	Net a year	Hardware payback
70	1,703	0.82	$42,242	$5,357	$36,885	0.8 months

- **Per call:** about $0.22 of running cost against about $1.65 of staff time, roughly 7.5 times cheaper.
- **Per resident:** a net saving of about $700 a year at 30 residents per robot.

Robots (base case)	One-time hardware	Net saving a year	Residents covered
1	$2,500	$20,917	30
10	$25,000	$209,170	300
50	$125,000	$1,045,850	1,500

The dashboard shows these live: calls resolved without a nurse, minutes saved with the assumption labelled, cost per call, and the share answered within 10 minutes.

Scope and guardrails

- **Prototype only.** Residents in the demo are teammates, there is no real patient data, and the medication tray holds fake pills.
- **The bell stays a bell.** Anyone can still press it to reach a person, and the robot never discourages someone from calling a nurse.
- **Lean toward escalation.** When the score is uncertain, Bellhop treats the call as more urgent, not less.
- **No diagnosis.** The robot gathers information and passes it on; it does not give medical advice.
- **Calm with confusion.** For residents with memory loss it stays gentle, never argues and never corrects them sharply.